

Experiment 6: Naïve Bayes Classification**Aim**

To implement the **Naïve Bayes algorithm** in Python and display the results using a **confusion matrix** and **accuracy**.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the dataset.

Step 3: Separate the features and target variable.

Step 4: Split the dataset into training and testing data.

Step 5: Create a Naïve Bayes classifier.

Step 6: Train the classifier using the training data.

Step 7: Predict the classes for the test data.

Step 8: Calculate the accuracy.

Step 9: Generate and display the confusion matrix.

Program

```
# Experiment 6 - Naive Bayes Classification

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

data = pd.read_csv("Iris.csv")

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Output

```
... Predictions: ['Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-virginica' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa']
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Result

Thus, the Naïve Bayes classification algorithm was successfully implemented in Python, and the classification performance was evaluated using accuracy and a confusion matrix.